

Safety data sheet

Page: 1/13

Chemetall (now part of BASF Group) BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 3.0

Product: **Gardolene ZL 6**

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

1. Identification

Product identifier

Gardolene ZL 6

Recommended use: Activating rinsing agent for metal surfaces before phosphating.

Not recommended use: Uses other than recommended

Details of the supplier of the safety data sheet

Company:
Chemetall GmbH
Trakehner Straße, 3
60487, Frankfurt am Main
Germany
+49(0)69 7165-0
sds.global-chemetall@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Eye Dam./Irrit. 2A
Aquatic Acute 3
Aquatic Chronic 3
Skin Sens. 1A

BASF Safety data sheet according to UN GHS 4th rev.
Date / Revised: 26.08.2021
Product: **Gardolene ZL 6**

Version: 3.0

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System, UN (GHS)

Pictogram:



Signal Word:
Warning

Hazard Statement:

H317	May cause an allergic skin reaction.
H319	Causes serious eye irritation.
H412	Harmful to aquatic life with long lasting effects.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P273	Avoid release to the environment.
P261	Avoid breathing dust/fume/gas/mist/vapours/spray.
P272	Contaminated work clothing should not be allowed out of the workplace.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P333 + P313	If skin irritation or rash occurs: Get medical attention.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P337 + P313	If eye irritation persists: Get medical attention.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P302 + P352	IF ON SKIN: Wash with plenty of soap and water.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.
------	---

Other hazards

According to UN GHS criteria

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 3.0

Product: **Gardolene ZL 6**

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

inorganic compounds, organic compounds, Water, inorganic salts

Hazardous ingredients (GHS)

According to UN GHS criteria

2-Methyl-2H-isothiazol-3-one

Content (W/W): > 0 % - < 0,1 %

CAS Number: 2682-20-4

EC-Number: 220-239-6

INDEX-Number: 613-326-00-9

Acute Tox. 2 (Inhalation - dust)

Acute Tox. 3 (oral)

Acute Tox. 3 (dermal)

Skin Corr./Irrit. 1B

Eye Dam./Irrit. 1

Skin Sens. 1A

Aquatic Acute 1

Aquatic Chronic 1

M-factor acute: 10

M-factor chronic: 1

H330, H317, H314, H301 + H311, H400, H410

EUH071

Specific concentration limit:

Skin Sens. 1A: >= 0,0015 %

Copper(II) carbonate--copper(II) hydroxide (1:1)

Content (W/W): >= 0,1 % - < 0,2 %

CAS Number: 12069-69-1

EC-Number: 235-113-6

INDEX-Number: 029-020-00-8

Acute Tox. 4 (oral)

Skin Corr./Irrit. 2

Eye Dam./Irrit. 2A

Aquatic Acute 1

Aquatic Chronic 1

M-factor acute: 10

M-factor chronic: 10

H319, H315, H302, H400, H410

Tetrapotassium pyrophosphate

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 3.0

Product: **Gardolene ZL 6**

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

Content (W/W): $\geq 15\%$ - $< 20\%$	Eye Dam./Irrit. 2A
CAS Number: 7320-34-5	H319
EC-Number: 230-785-7	
REACH registration number: 01-2119489369-18	

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Remove affected person from danger area. Immediately remove contaminated clothing. In all cases of doubt, or when symptoms persist, seek medical attention. Never give anything by mouth to an unconscious person.

If inhaled:

Remove the affected individual into fresh air and keep the person calm. If symptoms persist, seek medical advice. If breathing is irregular or stopped, administer artificial respiration.

On skin contact:

If symptoms persist, seek medical advice. Remove contaminated clothing. Wash skin with soap and water, rinse abundantly. Do NOT use solvents or thinners.

On contact with eyes:

Remove contact lenses, if present. Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist. Immediate medical attention required.

On ingestion:

Do not induce vomiting. Rinse mouth thoroughly with water, seek medical attention. If adverse health effects develop seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Eye irritation, allergic symptoms, Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Indication of any immediate medical attention and special treatment needed

Treatment: Symptomatic treatment (decontamination, vital functions).

Antidote: No known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:

BASF Safety data sheet according to UN GHS 4th rev.
Date / Revised: 26.08.2021
Product: **Gardolene ZL 6**

Version: 3.0

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

carbon dioxide, alcohol-resistant foam, dry powder, water spray

Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture

Hazardous decomposition products formed under fire conditions.

Advice for fire-fighters

Special protective equipment:
Appropriate breathing apparatus may be required.

Further information:

Cool closed containers in the vicinity of the source of fire. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. Product itself is non-combustible; fire extinguishing method of surrounding areas must be considered. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

Avoid breathing vapours. For non-emergency personnel: Use personal protective clothing. Ensure adequate ventilation. Keep away from sources of ignition. For emergency responders: Advice on product handling can be found in sections 7 and 8 of this safety data sheet. Information regarding personal protective measures, see section 8.

Environmental precautions

Do not allow to enter drains or waterways. Do not discharge into the subsoil/soil. If the product enters drains or sewers, the local water company should be contacted immediately; in the case of contamination of streams, rivers or lakes, the Environment Agency.

Methods and material for containment and cleaning up

Contain and collect spillage with non-combustible absorbent materials, e.g. sand, earth, vermiculite, diatomaceous earth and place in a suitable container for disposal according with the waste regulations (see section 13). Clean preferably with a detergent; avoid the use of solvents. Ensure adequate ventilation.

7. Handling and Storage

Precautions for safe handling

Provide good ventilation of working area (local exhaust ventilation if necessary). Do not return residues to the storage containers. Smoking, eating and drinking are forbidden in application area. For personal protection see section 8. Comply with the health and safety at work laws. Avoid inhalation of vapour and spray mist. The workplace should be equipped with an emergency shower and eye-rinsing facility. Avoid contact with the skin, eyes and clothing. Handle in accordance with good industrial hygiene and safety practice.

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 3.0

Product: **Gardolene ZL 6**

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

Protection against fire and explosion:

Avoid all sources of ignition: heat, sparks, open flame. The relevant fire protection measures should be noted.

Conditions for safe storage, including any incompatibilities

Keep away from oxidising agents, from strongly alkaline and strongly acid materials.

Suitable materials for containers: Polypropylene (PP), Polyethyleneterephthalate (PET), Low density polyethylene (LDPE), High density polyethylene (HDPE)

Further information on storage conditions: Keep container dry. Keep in a cool, well-ventilated place. Avoid direct sunlight. Close containers carefully once opened and store them upright in order to prevent any leakage. No smoking. No admission for unauthorised personnel. Always keep in containers of same material as the original one. Observe label precautions. Store protected against freezing.

Storage stability:

Storage temperature: 10 - 35 °C

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parametersComponents with occupational exposure limits

No occupational exposure limits known.

Exposure controlsPersonal protective equipment

Respiratory protection:

Respiratory protection required if exposure limit (if available) may be exceeded (Gas filter EN 14387 B)

Hand protection:

Chemical resistant protective gloves (EN ISO 374-1)

nitrile rubber (NBR) - 0.4 mm coating thickness

chloroprene rubber (CR) - 0.5 mm coating thickness

butyl rubber gloves - material thickness: 0.5 mm

Performance level 6, corresponding to a breakthrough time of >480 min according to EN ISO 374-1

The protection glove should be tested for its specific suitability (e.g. mechanical strength, product compatibility, anti-static properties).

The gloves should be replaced immediately in case of damage or signs of wear. It is recommended to use preventative skin protection (skin cream).

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 3.0

Product: **Gardolene ZL 6**

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

Eye protection:

Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:

Chemical resistant protective clothing according to DIN EN 13034 (Type 6)

General safety and hygiene measures

Do not breathe vapour/spray. Eye wash fountains and safety showers must be easily accessible. Avoid contact with the skin, eyes and clothing. Handle in accordance with good industrial hygiene and safety practice. Ensure adequate ventilation. This can be achieved by the use of local exhaust ventilation and good general extraction. If these are not sufficient to maintain concentrations at the workplace below the occupational exposure limits, appropriate certified respirators must be worn. Remove contaminated clothing immediately and dispose of safely. Hands and/or face should be washed before breaks and at the end of the shift. Keep separated from food stuffs and feed stocks.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	liquid
Colour:	blue
Odour:	perceptible
Odour threshold:	No data available.
pH value:	9,4 - 10,4 (10,00000 g/l)
Melting point:	< 0 °C
onset of boiling:	not determined
Flash point:	not applicable
Evaporation rate:	No data available.
Flammability:	not applicable
Lower explosion limit:	not determined
Upper explosion limit:	not applicable
Ignition temperature:	not determined
Vapour pressure:	(20 °C) not determined
Density:	(50 °C) not determined 1,240 g/cm ³ (20 °C)

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 3.0

Product: **Gardolene ZL 6**

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

Relative density: No data available.

Relative vapour density (air): No data available.

Solubility in water: completely miscible

Viscosity, dynamic: 70 mPa*s
(< 20 °C)
Rotovisco

Viscosity, kinematic: 6,0 mm²/s
(20 °C)

(40 °C)
not determined

Explosion hazard: not explosive

Fire promoting properties: not fire-propagating

Other information

Self heating ability: It is not a substance capable of spontaneous heating.

Miscibility with water:

Flow time: miscible
< 30 s

(DIN EN ISO 2431; 3 mm)

10. Stability and Reactivity**Reactivity**

No hazardous reactions if stored and handled as prescribed/indicated.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

No hazardous reactions when stored and handled according to instructions.

Conditions to avoid

Avoid direct sunlight. Avoid freezing.

Incompatible materials

Substances to avoid:

Keep away from highly acidic or alkaline substances as well as oxidants in order to prevent exothermal reactions.

Hazardous decomposition products

:

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 3.0

Product: **Gardolene ZL 6**

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

When exposed to high temperatures hazardous decomposition products such as carbon monoxide, carbon dioxide, smoke, oxides of nitrogen may be produced., No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:

Based on available data, the classification criteria are not met.

Irritation

Assessment of irritating effects:

Not irritating to the skin. Eye contact causes irritation.

Respiratory/Skin sensitization

Assessment of sensitization:

Sensitization after skin contact possible.

Germ cell mutagenicity

Assessment of mutagenicity:

Based on available data, the classification criteria are not met.

Carcinogenicity

Assessment of carcinogenicity:

Based on available data, the classification criteria are not met.

Reproductive toxicity

Assessment of reproduction toxicity:

Based on available data, the classification criteria are not met.

Developmental toxicity

Assessment of teratogenicity:

Based on available data, the classification criteria are not met.

Specific target organ toxicity (single exposure)

Assessment of STOT single:

Based on available data, the classification criteria are not met.

BASF Safety data sheet according to UN GHS 4th rev.
Date / Revised: 26.08.2021
Product: **Gardolene ZL 6**

Version: 3.0

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on available data, the classification criteria are not met.

Aspiration hazard

No aspiration hazard expected.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
Harmful to aquatic life. Harmful to aquatic life with long lasting effects. There are no test results available for this product. Do not allow to enter drains or waterways.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):
No data available concerning biodegradation and elimination.

Bioaccumulative potential

Bioaccumulation potential:
No data available.

Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: No data available.

Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

13. Disposal Considerations

Waste treatment methods

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 3.0

Product: **Gardolene ZL 6**

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

Observe national and local legal requirements.
No disposal via sewage or waste water systems.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport

ADN

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 3.0

Product: **Gardolene ZL 6**

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Transport in bulk according to Annex II of MARPOL and the IBC Code

Regulation:	Not evaluated
Shipment approved:	Not evaluated
Pollution name:	Not evaluated
Pollution category:	Not evaluated
Ship Type:	Not evaluated

15. Regulatory Information**Safety, health and environmental regulations/legislation specific for the substance or mixture**Not applicable

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 3.0

Product: **Gardolene ZL 6**

(ID no. 11077876/SDS_GEN_00/EN)

Date of print 11.03.2022

Eye Dam./Irrit.	Serious eye damage/eye irritation
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Skin Sens.	Skin sensitization
Acute Tox.	Acute toxicity
Skin Corr./Irrit.	Skin corrosion/irritation
H330	Fatal if inhaled.
H317	May cause an allergic skin reaction.
H314	Causes severe skin burns and eye damage.
H301 + H311	Toxic if swallowed or in contact with skin
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H302	Harmful if swallowed.
EUH071	Corrosive to the respiratory tract.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.